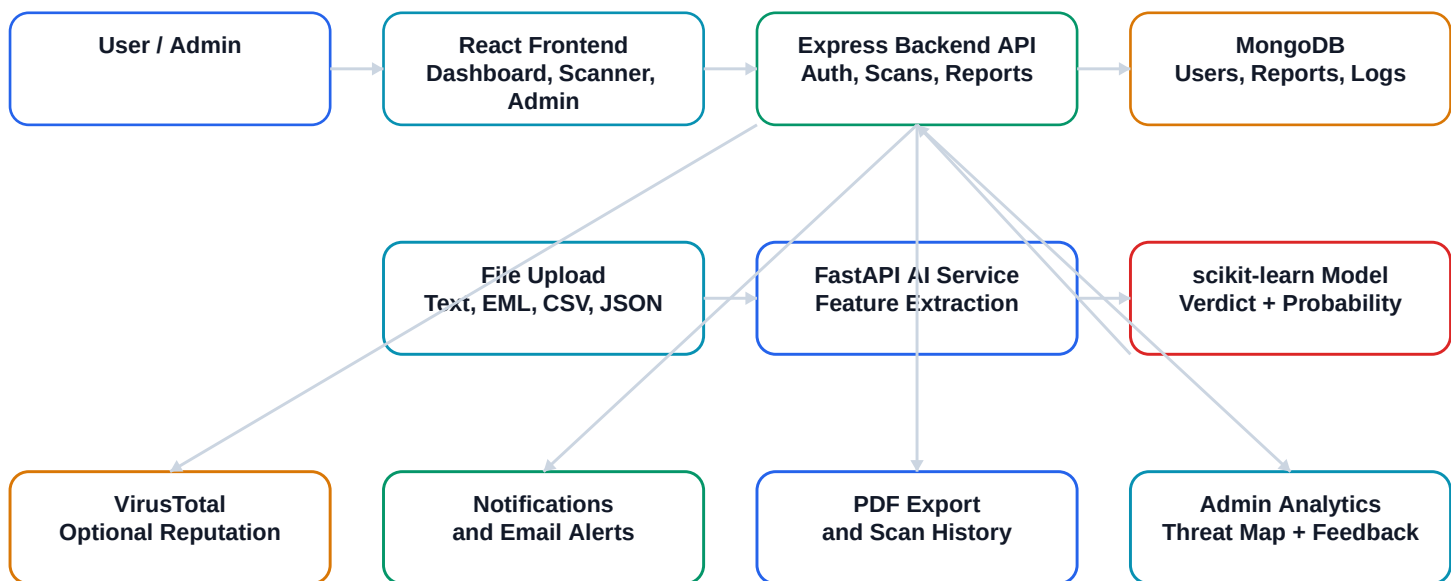


PhishGuard AI

Project Flow Chart and Technology Stack

This document summarizes the architecture flow and technologies used in the AI-based phishing detection website. It covers the React frontend, Express backend, FastAPI AI service, MongoDB data layer, testing tools, and deployment stack.

Project Flow Chart



Main Data Flow

1. User logs in or registers through the React frontend.
2. Frontend sends API requests to the Express backend.
3. Backend validates input, authenticates users with JWT, and stores records in MongoDB.
4. User submits a URL, text message, or uploaded file for phishing detection.
5. Backend forwards scan content to the FastAPI AI service.
6. AI service extracts features and returns probability, verdict indicators, and model details.
7. Backend optionally checks VirusTotal reputation signals for URL scans.
8. Frontend displays threat score, indicators, explainability details, history, and admin analytics.

Technology Stack and Uses

The following table lists the main technologies found in the project and explains how each one supports the phishing detection platform.

Technology	Used In	Purpose
React	Frontend	Builds login, dashboard, scanner, history, scan details, tips, and admin interfaces.
React Router DOM	Frontend	Controls protected routing and page navigation such as dashboard, scanner, history, and admin.
Vite	Frontend	Provides fast development server, React build pipeline, and optimized production assets.
Tailwind CSS	Frontend	Styles responsive layouts, dark mode, forms, cards, tables, and cybersecurity UI elements.
Lucide React	Frontend	Adds consistent icons for actions, alerts, navigation, and admin tools.
Recharts	Frontend	Renders dashboard charts, analytics, and visual scan summaries.
Axios	Frontend / Backend	Handles HTTP requests between frontend, backend, AI service, and optional external APIs.
React Hot Toast	Frontend	Displays user feedback for scans, uploads, exports, authentication, and model feedback.
Node.js	Backend	Runs the Express API server, scripts, and backend tooling.
Express.js	Backend	Provides REST routes for auth, scans, admin, notifications, chatbot, and PDF exports.
MongoDB	Database	Stores users, scan reports, threat logs, notifications, and feedback.
Mongoose	Backend	Defines schemas and performs database queries for MongoDB collections.
JWT	Backend	Protects routes using token-based authentication.
bcryptjs	Backend	Hashes user passwords securely before database storage.
Helmet	Backend	Adds security-focused HTTP response headers.
CORS	Backend	Allows the frontend app to communicate with the backend API.

express-rate-limit	Backend	Reduces abuse by limiting repeated API requests.
express-validator	Backend	Validates request payloads for auth, scans, filters, and feedback.
Multer	Backend	Processes uploaded scan files such as TXT, EML, CSV, and JSON.
Nodemailer	Backend	Sends email alerts when suspicious or phishing scans are detected.
PDFKit	Backend	Generates downloadable PDF scan reports and this project documentation PDF.
FastAPI	AI Service	Exposes prediction routes for URL and text phishing detection.
Uvicorn	AI Service	Runs the FastAPI server.
Pydantic	AI Service	Validates request and response schemas for AI endpoints.
scikit-learn	AI Service	Trains and runs the phishing detection machine learning model.
pandas	AI Service	Loads and processes training datasets.
NumPy	AI Service	Supports numerical operations for ML and data processing.
joblib	AI Service	Saves and loads trained model files.
VirusTotal API	Optional Service	Adds external reputation intelligence for URL scans.
Docker	Deployment	Containerizes frontend, backend, and AI service.
Docker Compose	Deployment	Runs the multi-service stack locally.
Vercel	Deployment	Hosts the production frontend.
Render	Deployment	Hosts backend and AI service using included configuration.
Vitest	Testing	Runs frontend and backend tests.

Testing Library	Frontend Testing	Tests React components from a user-focused perspective.
Supertest	Backend Testing	Tests Express API endpoints.
pytest	AI Testing	Tests Python feature extraction and prediction behavior.

Summary

PhishGuard AI is a full-stack phishing detection platform. The frontend provides the user experience, the backend handles secure APIs and persistence, the AI service performs model-based detection, and MongoDB stores application data. Testing and deployment tools make the project easier to verify, run, and host.